

A Review of A Multivariate Approach to the Study of Orichalcum ingots from the Underwater Gela's Archaeological Site

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1. Abstract

General Background

A shipwreck off the coast of Gela, Italy has been the source of a historical find of 40 orichalcum ingots. A scarce, fabled metal that has the modern equivalent of brass.

sPECIFIC BACKGROUND

Knowledge gap

Large groupings and finds from one source are very hard to come by: lots of data in one place.

Here we show...

After performing a trilogy of cluster analyses, we show that were the clusters accurate were we able to determine a good fit?

Our goal is to create and present a comprehensive, comprehensible, reproducible, and replicable multivariate analysis of the data obtained from the chemometric elemental analysis of 38 orichalcum ingots that were discovered in a 6th Century BC shipwreck off the coast Gela, Italy in 2015.

Results

Implications

They had different origins. Can possibly be linked to geological locations by ore analysis, collected from different regions, etc.

Why did they wait to do further testing that would provide more information on the composition and provide more accurate data?

The groupings from the Caponetti, Armetta, Brusca, et al. (2017) study appears to have been manipulated without compensation of the . By taking the log at any point in this anylysis without standardisation would have resulted in multiple zero-values and artificially inflating

Is the cluster analysis consistent with different distances and linkages?

This is the largest sample of its kind however it is still not a large sample.

For the analysis of our report we will be considering all the main components and trace element concentrations excluding the Cu/Zn ratio as...

Why do we care about the problem and the results?

Because it could lead to further impressive archaeological finds

What is the scope of the work and what problem is this paper trying to solve?

Determine whether or not any of the ingots could be grouped. The scope is mapping orichalcum origin.

What was done to solve this problem?

More destructive testing to determine trace elements.

What implications does the answer imply?

one of our hypotheses questions is whether our data follows a multivariate normal distribution. probably some hypotheses around the clusterings.

2. Introduction

2.1 Significance of this archaeological find and Gela

The historic Gela, Italy has been a treasure trove of archaeological finds for generations. This ancient city was strategically positioned at the crux of major maritime trade routes, which made it a populous and successful trading port (Angelucci, 2026). Consequently, ships looking to trade goods like pottery, metals, and grain have traveled to and from Gela for centuries.

A shipwreck off the coast continues to provide significant archaeological finds to this day. This discovery dates back to 1988 (Nutter, 2025), where it was first stumbled upon by divers near Bulala beach. We are exploring one of the most significant finds from this wreck; A veritable hoard of the fabled metal, Orichalcum.

The discovery of 40 orichalcum ingots in 2017 was a monumental one. There had never been a discovery of this quantity in one place anywhere in the world. There was also evidence of consistent casting processes, with overall similarities suggesting a rudimentary mold production. This was likely a hole dug in the ground and molten alloy was poured in the indentation to solidify. The morphology of the found ingots suggested cast-diverse technologies.

2.2 What is orichalcum?

Orichalcum is defined as a copper alloy with zinc content as the principal components. A widely varying quantity of lead and tin among other trace elements are expected. There is no consistent proportion to the amount of copper and zinc found, but the overall composition and relative amounts of principal elemental components can be used to deduce the rough date and origin of a sample. The modern day equivalent is brass.

2.3 Summary of Collection Techniques

In the initial analysis of the orichalcum ingots, X-Ray Fluorescence (XRF) spectroscopy via a portable spectrometer (Caponetti, Armetta, Martino, et al., 2017) was used. The subject paper includes that data and further investigates the composition of each ingot using more invasive and refined elemental analysis.

To obtain material for further testing, a micro-well was drilled into each ingot and the extracted alloy was taken to determine trace elements using Inductively coupled plasma optical emission spectroscopy (ICP-OES) and Inductively coupled plasma optical emission spectroscopy (ICP-MS) of each of the 38 orichalcum ingots. (FLESH OUT AND EXPLAIN MORE, TOO VAGUE NOT EXPLAINING WHY)

2.4 Scope of This Multivariate Analysis

This multivariate analysis includes manipulations of the data collected from the ICP-OES and ICP-MS chemometric analysis, it does not include the data about physical characteristics of the ingots or any correlation analysis of those variables.

We will be excluding the Cu/Zn ratio variable as we chose to analyse the linkage of all elements between each other, not between Cu/Zn ratios and other elements

3. EDA

3.1 Analysing raw data

We performed a simple summary of the data in table 1 and creating a box plot to visualize the data. Something to note is that the elements Cu, Zn, Pb, and Fe are all in percentages, where as the concentrations of the other elements are in PPM (parts per a million). This immediately indicated that scaling was needed.

Table 1: Figure 1: Main component and trace element concentrations obtained by ICP-OES & ICP-MS techniques

Sample	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
S1	73	21	5.6	0.30	580	146.0	55.7	9.85	2.79	12.2	9.70	0.57	41.4	19.9	5.55	11.50	3.71	3.17	0.08	1.00	0.19
S2	87	16	4.2	0.53	534	199.0	32.0	36.4	08.44	20.0	4.80	1.00	72.8	20.8	2.17	4.46	1.55	1.31	0.07	0.15	0.18
S3	73	28	5.9	1.07	845	131.0	105.0	23.8	08.47	16.5	6.70	0.64	29.0	15.7	4.00	8.89	3.33	2.70	0.14	8.90	0.35
S4	80	20	6.1	0.39	558	171.0	70.0	11.1	04.83	11.9	9.34	0.82	60.7	23.2	6.16	13.80	6.08	4.91	0.14	0.29	0.39
S5	78	22	3.8	0.55	513	156.0	67.0	14.5	06.49	5.3	7.69	0.58	58.7	23.3	3.87	11.60	4.38	3.69	0.12	0.91	0.45
S6	81	22	6.5	1.0	957	159.0	60.3	8.32	1.88	9.6	8.54	0.96	61.4	15.7	2.12	4.36	2.29	1.90	0.11	0.45	0.75

Sam-

ple	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
S7	75	20	6.45	0.33	559	130.0	07.0	211.0	0.37	11.6	17.6	0.80	51.0	22.3	1.75	5.50	2.16	1.57	0.13	0.34	0.50
S8	82	24	6.13	0.31	548	160.0	75.4	9.66	3.19	7.6	6.98	0.83	58.5	23.7	1.99	13.3	0.55	2.36	0.11	1.45	0.26
S9	80	18	4.63	0.44	482	159.0	09.0	24.2	05.28	30.5	3.92	0.91	91.3	23.7	1.69	8.40	2.70	1.78	0.07	0.11	0.43
S10	72	28	5.92	0.76	435	105.0	11.0	23.5	08.29	16.2	6.65	0.61	27.6	19.5	1.23	6.77	2.00	1.41	0.12	0.21	0.30
S11	76	19	5.93	0.43	525	138.0	3.2	15.1	05.13	8.5	8.29	0.86	54.5	12.9	0.92	6.57	1.71	1.28	0.10	0.56	0.24
S12	69	17	6.67	0.50	657	141.0	0.1	13.8	04.45	8.3	8.14	0.87	55.0	13.4	1.20	3.22	1.65	1.10	0.10	3.13	0.22
S13	78	20	3.30	1.06	458	189.0	49.0	97.0	0.60	17.3	38.3	0.00	181.0	86.6	1.55	9.28	3.08	2.23	0.07	0.52	1.17
S14	74	25	4.70	0.44	433	81.7	88.2	14.6	05.08	20.3	1.98	0.71	26.1	29.7	1.13	6.51	2.49	1.55	0.10	0.17	0.19
S15	84	20	5.20	0.14	549	171.0	07.0	20.6	01.06	23.4	4.24	0.95	92.0	12.0	0.74	3.44	1.58	1.08	0.08	0.08	0.17
S16	82	19	4.72	0.16	560	162.0	03.0	9.90	1.14	22.2	4.00	0.89	82.1	12.4	0.67	21.2	0.32	2.53	0.08	0.31	0.31
S17	77	20	6.92	0.79	500	177.0	153.0	4.70	9.06	31.8	65.8	0.81	27.9	5.8	0.40	3.31	0.93	0.67	0.08	7.27	0.35
S18	83	19	4.57	0.44	515	158.0	11.0	22.1	05.09	28.4	3.80	0.99	89.3	21.5	0.52	4.08	1.69	1.01	0.09	11.8	0.27
S19	71	28	5.25	0.72	494	114.0	01.0	8.8	106.14	18.5	10.4	0.72	33.6	17.7	1.49	8.36	2.61	1.54	0.16	4.89	3.37
S20	76	19	6.31	0.41	535	139.0	1.5	14.0	04.41	8.2	8.11	0.97	56.3	13.4	0.62	2.91	1.34	0.79	0.06	0.42	0.40
S21	74	21	4.88	0.49	423	80.0	68.0	14.4	05.67	22.5	1.99	0.91	25.3	26.3	0.42	2.49	0.97	0.59	0.07	0.70	0.23
S22	71	21	6.00	0.93	606	142.0	175.0	5.30	15.03	11.8	42.0	0.56	46.3	25.7	0.85	11.6	0.82	2.19	0.12	15.1	0.61
S23	72	22	6.18	0.33	623	121.0	67.2	9.12	2.61	11.6	9.07	0.58	40.1	18.3	0.73	5.38	2.52	1.45	0.11	0.08	0.14
S24	76	18	6.49	0.28	645	172.0	52.5	9.43	1.94	9.1	9.75	1.00	79.7	14.6	0.40	3.76	1.39	0.86	0.08	0.15	0.28
S25	78	18	1.05	0.75	496	162.0	2.2	56.7	00.34	5.0	4.10	0.91	49.8	29.5	0.61	5.31	2.02	1.27	0.09	0.17	0.32
S26	75	20	6.35	0.45	543	137.0	57.1	9.50	4.95	11.2	8.29	0.85	58.6	23.4	0.49	14.6	0.69	1.10	0.07	0.19	0.25
S27	72	20	6.48	0.41	558	140.0	54.5	9.10	4.34	10.6	8.19	0.89	57.0	18.4	0.36	3.05	1.49	0.85	0.07	6.73	0.26
S28	76	21	5.91	0.83	471	172.0	139.0	46.5	08.88	54.7	1.89	0.66	27.0	12.5	0.41	6.32	2.03	1.60	0.09	4.00	4.59
S29	69	28	6.57	0.81	492	121.0	22.0	26.4	07.79	18.8	9.63	0.71	33.3	22.1	0.53	5.25	2.74	1.81	0.13	0.93	0.38
S30	86	16	4.61	0.57	467	196.0	158.0	7.70	10.40	19.8	4.90	0.99	74.2	22.5	0.19	2.16	1.13	0.67	0.07	0.28	0.20

found that S22, S18, and S17 appear to have high Sr content (15.1ppm, 11.8ppm, 7.27ppm respectively compared to a median of 0.42ppm) while S28, S22, and S19 have high Ba content (4.59ppm, 3.61ppm, 3.37ppm respectively compared to a median of 0.28ppm).

(MAYBE SAY 'further data clustering analysis may find that these groupings of samples cluster together in similar groups, furthering Caponetti, Armetta, Brusca, et al. (2017) claim that trace element analysis could be used to differentiate providence of alloy production)

Figure 2: Element Concentration Spread of 38 Orichalcum Ingots

38 Orichalcum Ingots | Notched Boxplots | Highlighted Outliers

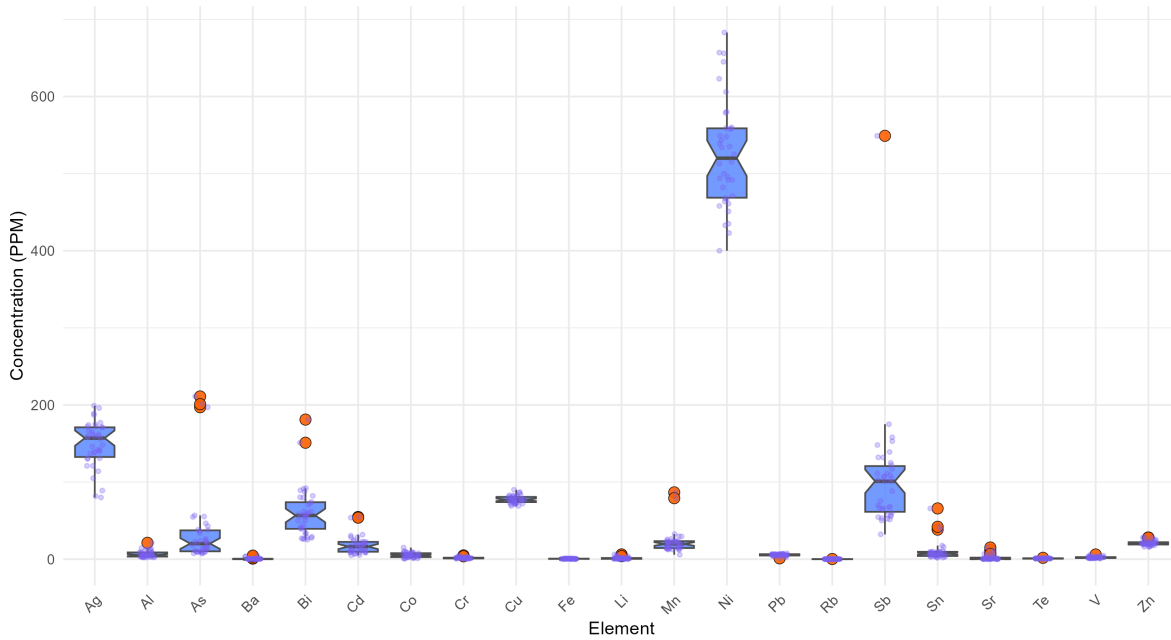


Figure 1: Notch Boxplot

Ni (nickel) had the widest interquartile range, indicating the greatest variation among the samples, while Sb (antimony) shows the largest total range, likely driven by sample 13's recorded Sb being magnitudes larger than any of the other samples.

Due to the fact the data was recorded at different scales (percentage and PPM) and obvious outliers, log transformation and scaling were the next logical steps.

3.1.1 Inconsistencies in Percentages

The percentages of elements Cu, Zn, Pb, and Fe do not add up to 100% for each sample, as shown in the table below.

Trace elements weren't included for interpretability as the values are $\times 10^{-4}$ smaller.

Table 2: Main Component Elements concentrations per Sample

Sample	Cu	Zn	Pb	Fe	percentage_total
S1	73	21	5.60	0.30	99.90
S2	87	16	4.29	0.53	107.82
S3	73	28	5.91	0.78	107.69
S4	80	20	6.17	0.39	106.56
S5	78	22	3.87	0.55	104.42
S6	81	22	6.51	0.19	109.70
S7	75	20	6.45	0.33	101.78
S8	82	24	6.13	0.31	112.44
S9	80	18	4.63	0.44	103.07
S10	72	28	5.92	0.76	106.68
S11	76	19	5.93	0.43	101.36
S12	69	17	6.67	0.50	93.17
S13	78	20	3.30	1.06	102.36
S14	74	25	4.70	0.44	104.14
S15	84	20	5.20	0.14	109.34
S16	82	19	4.72	0.16	105.88
S17	77	20	6.92	0.79	104.71
S18	83	19	4.57	0.44	107.01

Sample	Cu	Zn	Pb	Fe	percentage_total
S19	71	28	5.25	0.72	104.97
S20	76	19	6.31	0.41	101.72
S21	74	21	4.88	0.49	100.37
S22	71	21	6.00	0.93	98.93
S23	72	22	6.18	0.33	100.51
S24	76	18	6.49	0.28	100.77
S25	78	18	1.05	0.75	97.80
S26	75	20	6.35	0.45	101.80
S27	72	20	6.48	0.41	98.89
S28	76	21	5.91	0.83	103.74
S29	69	28	6.57	0.81	104.38
S30	86	16	4.61	0.57	107.18
S31	77	25	5.25	0.73	107.98
S32	82	22	7.22	0.72	111.94
S33	82	20	5.14	0.45	107.59
S34	80	16	4.48	0.47	100.95
S35	78	19	6.62	0.24	103.86
S36	78	20	7.71	0.18	105.89
S37	75	23	6.63	0.23	104.86
S38	90	21	2.33	0.38	113.71

Table 3: Data summary

Name	main_com[, 6]
Number of rows	38

Number of columns	1
Column type frequency:	
numeric	1
Group variables	None

Variable type: numeric

skim_variable	mean	sd	p0	p25	p50	p75	p100	hist
data	104.36	4.28	93.17	101.45	104.4	107.14	113.71	

Table 4: Summary statistics for main component elements.

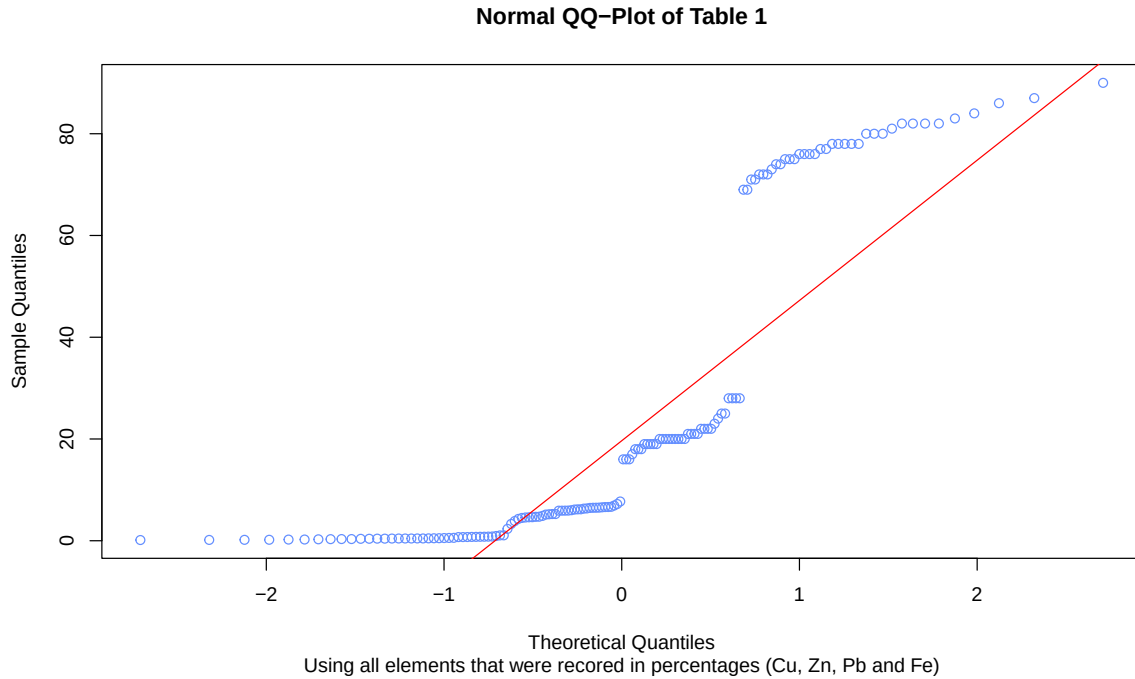
Element	Mean	SD	Median	Min	Max
Cu	77.42	4.98	77.00	69.00	90.00
Zn	20.95	3.25	20.00	16.00	28.00
Pb	5.50	1.34	5.92	1.05	7.71
Fe	0.50	0.23	0.44	0.14	1.06
percentage_total	104.36	4.28	104.40	93.17	113.71

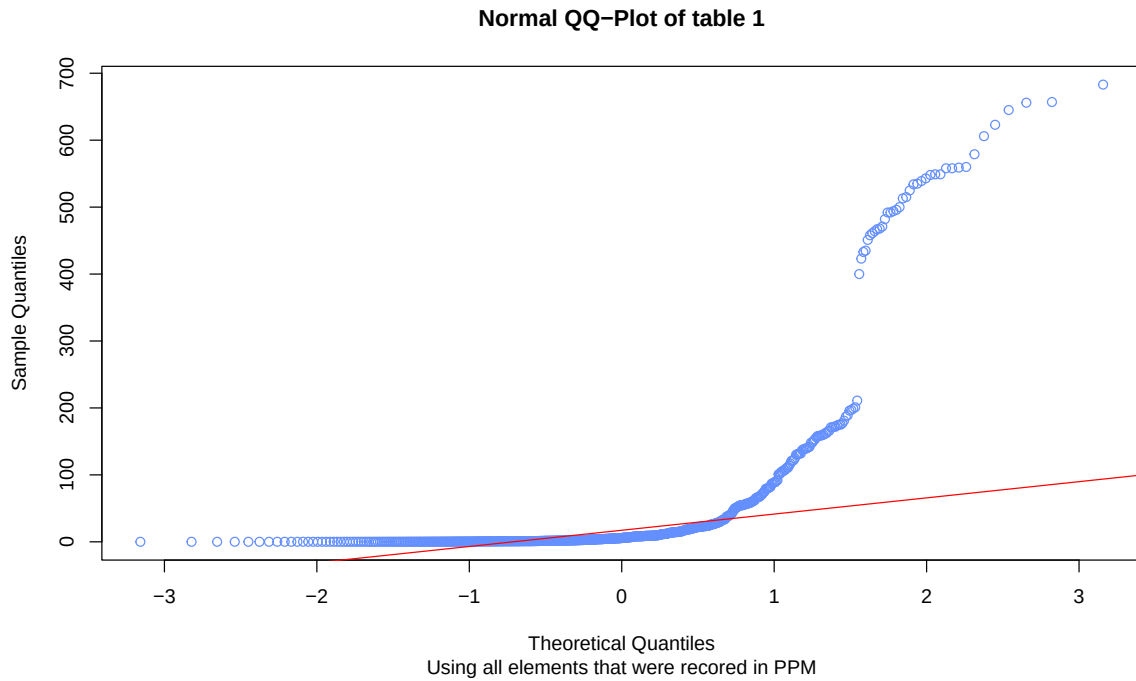
These samples have a mean of 104.4\% with a range 93.17\% to 113.71\%. Where 33 of the 38 samples exceed 100\%. The most significant deviations are samples 8 and 38 (COMMENT ON THIS). Caponetti, Armetta, Brusca, et al. (2017) attribute this to be due to errors generated from the low quantity of available material for sampling for each observation.

These samples have a mean of 104.4\% with a range 93.17\% to 113.71\%. Where 33 of the 38 samples exceed 100\%. The most significant deviations are samples 8 and 38 which have percentage totals of 112.44\% and 113.71\% respectively. Caponetti, Armetta, Brusca, et al. (2017) attribute this to be due to errors generated from the low quantity of available material for sampling for each observation. For example, only 12.3 mg of sample 38 were obtained compared to the ideal 20mg of material as described in their methods.

3.2 QQ-plots & Henze-Zirkler Test For Multivariate Normality

We first compared the QQ-normal plots of table 1 and compared it to the log transformed table 1 to assess if log transformation would be appropriate to manage outliers and standardisation.





As this was before there was any scaling performed, we opted to display the elements measured in percentages and the elements measured in PPM separately to better display their individual influence.

Both Percentages and PPM show blatant departures from the assumption of normality. Few of the points follow the theoretical model line, steps/gaps in the points are likely because the various element concentrations have such different numerical ranges, both plots somewhat show an s-shaped curve which tend to indicate that the variables have heavy skew (extreme values/outliers) which is unsurprising considering the ranges and variations between elements being widely different. Overall significant departures from normality can be seen for both Percentages and PPM. We do note that the qq-plot for percentages is marginally better due to the more symmetric distribution of values.

```
#log transform data

logIngots <- log(as.matrix(Ingots_F_no[-1,-1]))

#print(logIngots)

knitr::kable(logIngots,
              caption = "Log Transformed Data",
              format = "pipe")
```

Table 5: Log Transformed Data

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
2	4.462978	4.154856	2.867628	6.326326	3.385285	2.563225	2.993175	2.860600	0.080703	0.973474	3.147314	4.882379	0.271	-	-	-	-	-	-	-	-
			0.6348783																2.659289	7.1200	4.7984
3	4.290433	4.220766	4.586114	4.874167	3.966268	6.580330	0.210753	3.622053	1.638629	4.492709	0.993251	8.186051	3								
			0.2484614							0.4462871									1.966113	1.0498221	
4	4.382027	4.178296	9.886324	5.416342	4.050947	4.465234	3.063110	3.914418	1.280608	1.686500	0.912739	-	-								
			0.9416085							0.1984509									1.966123	7.8744	1.6085
5	4.356709	4.104523	2.545624	6.276982	6.266934	1.890263	2.703992	0.810734	1.084332	1.451007	7.038562	65	-	-							
			0.5978370							0.5447272									2.120209	4.3708	5.077
6	4.394409	4.108233	3.956530	6.909233	2.866312	2.632763	4.761011	2.475067	1.146714	1.828554	1.8539	-	-								
			1.6607312							0.0408220									2.207279	8.5087	6.821
7	4.312488	4.178240	8.016326	4.867567	2.829183	4.213206	6.789893	3.826058	5.796708	7.780148	2.0756	-	-								
			1.1086626							0.2231436									2.040207	8.8093	1.472
8	4.406719	4.085431	9.476306	2.751172	2.826719	6.020281	9.430489	1.069027	5.465821	3.467040	0.856661	6.3715636									
			1.1711830							0.1863296									2.207275	1.3470736	

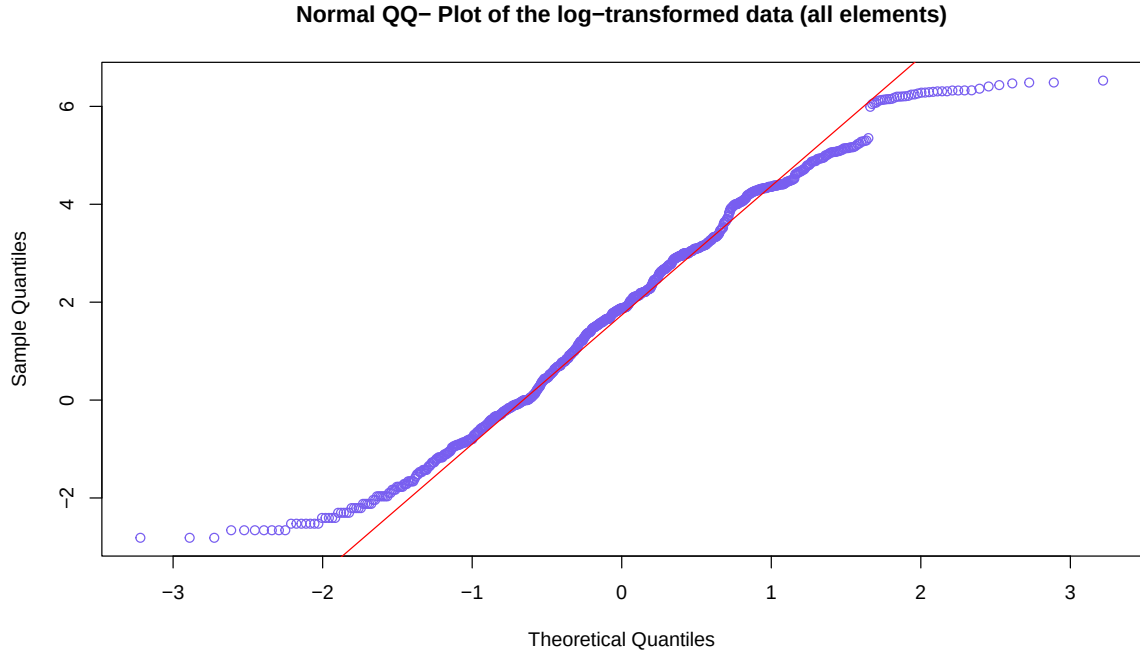
	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
9	4.3820870352	2.55666	1.779044896	1.341861363	2.411736760	917.51415154	472.472882997	25786134	-	-											
		0.8209806									0.0943107								2.65922072849	701	
10	4.276633227	83364.0743453	19609585200052	786108946163	3128976042071	94256911373	5897	-	-												
		0.2744368									0.4942963								2.12025406207	9728	
11	4.332793444	73002425.26339282974	205846655105120665	05003.9922517227	1.88251361246	8601	-	-													
		0.8439701									0.1508229		0.0833816						2.30255798485	1164	
12	4.2341873283	76196.4876848761	20224649220422596	79024.002333625823	16605007093	31021.1410330															
		0.6931472									0.1392621								2.302585	1.5141277	
13	4.3567991782	1902382185829673	47802832470208367665	44090009849703408222986249	800016	-	0.1570037														
																			2.659226039265		
14	4.3040258857	6756256.07473834572	66810622530106683	30968.26393501172287639932838	2549	-	-														
		0.8209806									0.3424903								2.302585195668	7312	
15	4.4328971762	286586.308094166728229529182	689734456334.5227894907	1.2350475571078	9610	-	-														
		1.9661129									0.0512933		0.3011051						2.52573257736	9568	
16	4.4067944439	180886.327938756347290729102800392	62944.4029387697	3.054084156282493	-	-															
		1.8325815									0.1165338		0.4004776						2.525729111830	1830	
17	4.3428951732	44158.214608616394382824384594468	66198.3286277858	1.1969482	-	-	1.9837563														
		0.2357223									0.2107210		0.9162907	0.0725740024526	729	1.0498221					
18	4.4189444451	9951335.2451067259953915678723748335	50014.4920068053	1.40609707289	95032.4680995																
		0.8209806									0.0100503		0.6539265						2.407946	1.3093333	
19	4.2626332265	822816.20257341683124012814894	273418958.51252656987	763495893462	78241.587112249127																
		0.3285041									0.3285041								1.832582		
20	4.33279344439	213576.282263747425829048387021094	309794.0306995255	1.0681392	16696	-	-	-													
		0.8915981									0.0304592		0.4780358		0.2357323418675	9062907					

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
21	4.304004453251453	0.047378240272568127351393568813463.23082995690.9122827	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.7133499										0.0943107		0.8675006	0.0301522563728666496760							
22	4.262680452217593	0.406885582647862778024683103766963.83312264912.45110030173699012.71462837078																			
	0.0725707										0.5798185		0.1625189						2.120264		
23	4.276606108213183	0.4345763720267304759346202649723.69237069011.682682423895636	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	1.1086626										0.5447272		0.3147107						2.202253579861129		
24	4.3307890387202623	0.4692507496082438862688227472070007026810211.324013903937	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	1.2729657												0.9162907		0.15083221789712009657						
25	4.356789037287903	0.20650783597190377741.60944809873.90803843901.669570802790169	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.2876821								1.0788097		0.0943107		0.4942963						2.407976193684343		
26	4.312488173284548	0.2971099498448041259238476911505004.07673527362.681021470953102	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.7985077										0.1625189		0.7133499						2.659266073882944		
27	4.2766061736287203	0.3243548649220824678762350291394.04309123511.11513987761	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.9065751		
	0.8915981										0.1165338		1.0216512		0.16256592601.3470736						
28	4.3307890387202623	0.4692507496082438862688227472070007026810211.324013903937	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.1863296										0.4155154		0.8915981						2.407946		
29	4.234137228525138	0.198479317904027330328909326748833.503559755781.65820819593268	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.2107210										0.3424903		0.6348783						2.040207259675840		
30	4.45237725822279	0.146329810625929630189886582923524.30676135150.77010822476	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.5621189										0.0100503		1.6607312		0.40021656280296594379						
31	4.343805887682285	0.9914688665212023300583195612563.26956943121.376238078087902	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.3147107										0.3856625		1.1711830						2.302586614291164		
32	4.40670910426855	0.13398729272726282794820438043.66612390571.556097028282549	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0.3285041										0.0512933		0.9416085						2.120261479841830		

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
33	4.406799517632705316.19847949862172476601.3109278160000704981910	0.7985077												1.9661129		0.5276827410866305815					
34	4.382077154996236.1484231189286224633293685078502080201928523	0.7550226												1.9661129	0.11053821889440794607312						
35	4.356799448900954.28971590552019946472.032028033951.382767228	1.4271164									0.0100503		1.7719568		0.328562117297117649568						
36	4.356799270225183.52649647871306068866162882500600021633242	1.7147984											1.2039728		2.207286750006760						
37	4.317483149116048.139488711989655622082007312263313.7037681295	1.4696760									0.4307829		2.3025851		2.407946	2.1202635					
38	4.493804052258683.486160498283343305	0.9675840								0.1053605				0.9162907		1.514128	0.8675006				

```
#taking the log.
```

```
qqnorm(logIngots, col = "#785EF0", main = "Normal QQ- Plot of the log-transformed data (all e
qqline(logIngots, col = "red")
```



The QQ-plot of the log transformed data roughly followed the theoretical line (in red), this is evidence that the normality assumption may hold. The tails are relatively long and flat, indicating that there may be some extreme outliers. There are several steps in the upper and lower tails, which alone is not necessarily a departure from normality, and may be explained by the zero values returned for some Sample's variables after the log transformation.

The observed zero-values for (PUT SAMPLES NUMBERS IN HERE) in (TABLE) in variables(HERE), which while not a large quantity of the samples, still provided some concern as $\log(0)$ is undefined and can influence non-normality in multivariate data. The negative values are not an issue because we will be using metrics which are calculated from coordinate distances for clustering, and can handle negative values.

We chose to combat the zero values by standardising the data set using $\log(x + 1)$, where x is our data. This, coupled with a Henze-Zirkler test, established if there is preliminary evidence

for multivariate normality.

```
#log transform data

logIngots1 <- log(as.matrix(Ingots_F_no[-1,-1] + 1))

#print(logIngots)

knitr::kable(logIngots1,
              caption = "Log Transformed Data",
              format = "pipe")
```

Table 6: Log Transformed Data

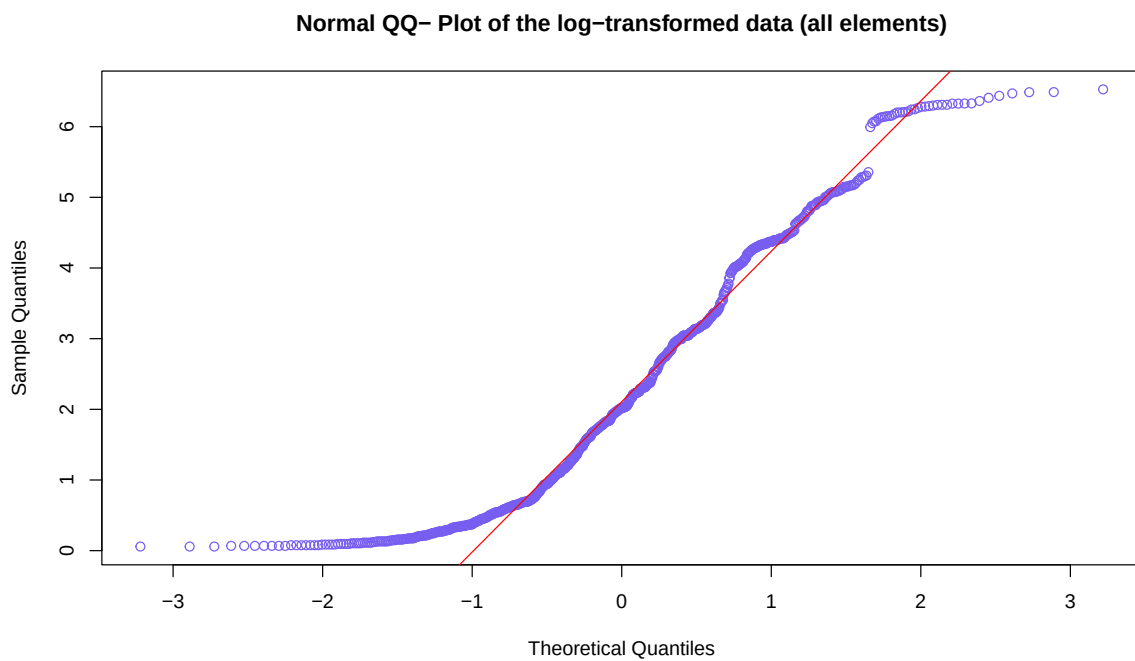
	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
2	4.472387	326638	1832087	2207838	9703692	6744956	6152378	5831372	3591915	3789644	9608372	0676580	7669514	4							
3	4.304066	729629	676613	682286	342928	448128	220412	294696	218734	609213	915265	678333	802935	308104	6						
4	4.394404	452090	637303	514974	294264	932063	015702	376059	830328	863568	569462	772730	643802	836322	3037						
5	4.363418	549830	938234	225624	624625	782840	356825	562173	740883	320458	309396	982688	413268	087103	25636						
6	4.406719	524946	235393	630085	174527	821637	790083	345947	204335	855409	788389	690806	071073	670553	96158						
7	4.336703	445008	228613	890873	108731	356586	476306	973015	873967	214846	316809	805057	200039	227066	064651						
8	4.418821	889643	270030	809844	343298	664982	706762	696843	066976	680932	684259	512719	400869	008801	117						
9	4.392494	447381	094648	007517	034826	883737	009588	336971	632024	680895	240730	830284	609658	036674	4						
10	4.296436	729641	553073	660347	981998	673298	880909	070662	343407	048620	056209	868796	263329	062623	643						
11	4.342809	579338	598676	430344	942687	883132	124722	228962	050662	363086	523030	193698	010943	402685	8114						
12	4.242489	023737	346646	829558	273769	460756	230021	266250	083360	722884	439835	455900	095340	82798	509						
13	4.363404	452386	730700	002070	399028	829535	906967	102053	120200	727830	934200	609704	087638	677037	272						

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
14	4.31742880970166260330444522028472805004870970923619935324026361026226990370099316700339533																				
15	4.4426644582454930389018740821672672270045636366783925594094538890694778923670067006570037																				
16	4.418899573230688430670937604393977608054162946365708045525528206092990682070007002700271																				
17	4.3567004502939822216608578469532930856704290759333638892693364700968752008036961063401046																				
18	4.4308975732789646246106890434999886363809568668813061383545876033989569810862157942390169																				
19	4.27663672862584332035784962406766126379724433653233438928523228374253707381648427024607630																				
20	4.342809579892333588719436902870806882292209036780038306722824263585015892056268063604722																				
21	4.31748810721530870407394429210743873619970095263710625696883506249962803837060658062870142																				
22	4.27660010923066752085292817644832675216234761200468585284664521536697025600209327885282279																				
23	4.293459549712295178618040222434528376786979565734748006015481854128846900880307006300283																				
24	4.34280424693508886070805329296824686821092537496631392773972364500283120305066963970408601																				
25	4.3624944079783986268599375025655239267907692024731032896772762842136328687086175700276317																				
26	4.33670345224707150969927264218513783290242889695185665645898276712789574119067658692331436																				
27	4.293464450222328589619987602333253522252025816865066496523674898797226271866668610891117																				
28	4.3428091093206063166079329236802720050299860257683372266269035890610850365086160747209793																				
29	4.248496220941993320858940812181251735982683658619331139832526372582908361823265752000835																				
30	4.4629883273454670746408320489645848360349574968813261557000739550572612208066638686823216																				
31	4.3567098087258451294969986029762239607642345845987908887452600071402967978098310702831114																				
32	4.41881324906550232335663906397626307500669054607809423768032937374235569600933328752700271																				
33	4.4188044582483775006509797068696854188435923860314828726763102832673146370348208917384200																				
34	4.392489327010812634623648923449261521086680837830968298061910286863657680056208017739533																				
35	4.3624982703072641294566478623364566822925357698813961764365770089283724733076901011570037																				
36	4.3634084520417685512705804886608442929000480409331472132029623638986000885600866063690142																				
37	4.33673780642080780142088282046931124202378284009607732108500953302980207060086135783333287																				

38 4.510859102297230485684119886280826818230237454091600988832736272258932069708856913686569

```
#taking the log.
```

```
qqnorm(logIngots1, col = "#785EF0", main = "Normal QQ- Plot of the log-transformed data (all  
qqline(logIngots1, col = "red")
```



```
HZoriginal <- mvn(data = logIngots1, mvn_test = "hz")  
cat("Henze_Zirkler Test Results:\n")
```

Henze_Zirkler Test Results:

```
print(HZoriginal$multivariate_normality)
```

	Test Statistic	p.value	Method	MVN
1 Henze-Zirkler	1	0.069	asymptotic	Normal

As you can see in table 2, our values are now all non-zero.

The normal QQ-plot looks less normal after standardisation, with the length of the tails being accentuated and continuing presence of stepped data.

The Henze-Zirkler test stipulates

$\mathcal{H}_0$: The data follow a multivariate normal distribution,

$\mathcal{H}_1$: The data do not follow a multivariate normal distribution.

This generated a p-value=0.069, suggesting that there is insufficient evidence to reject that the data follow a multivariate normal distribution. However, the QQ-plot shows enough <flat tails, step, steep, etc> that further testing is warranted before determining if multivariate normality is likely.

Given the uncertainty of multivariate normality and the presence of outliers, further data manipulation by Mediod Absolute Deviation was determined to be the next best step towards multivariate analysis.

3.3 MAD

MAD was used because it is a robust, outlier-resistant, measure of variability. The log transformation somewhat fixed the skew and pulled the outliers in, but did not eliminate their mathematical influence.

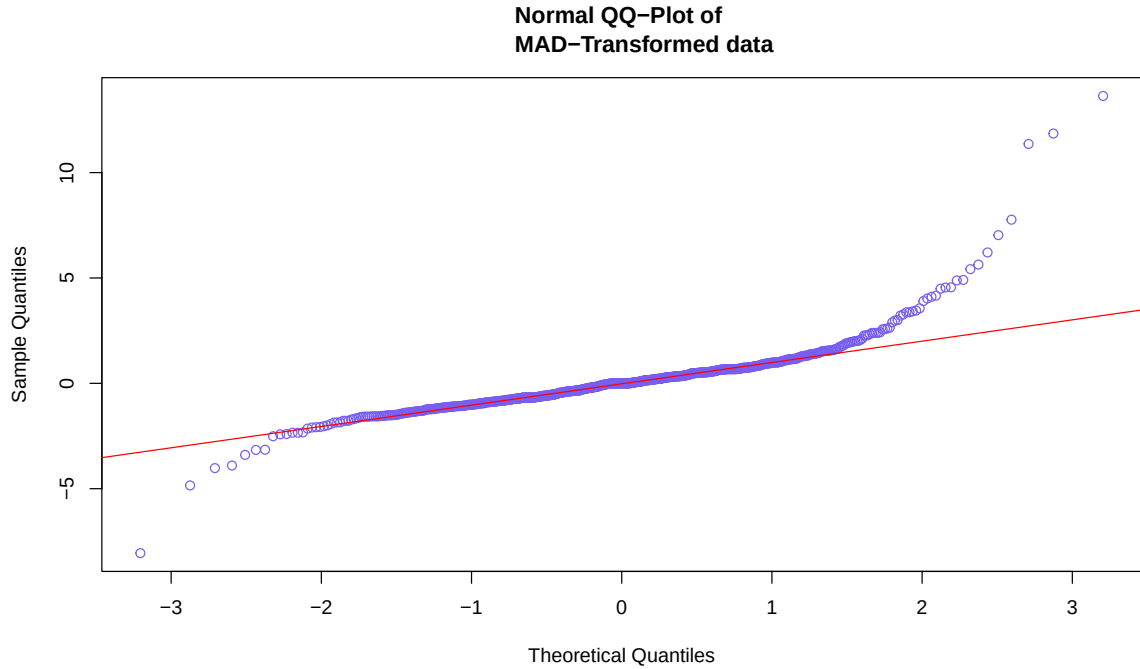
Table 7: Medians / Mads

	Median	Mad	Median_over_Mad
Zn	3.0445224	0.1348748	22.572959
Pb	1.9344158	0.1509795	12.812436
Fe	0.3715636	0.1505379	2.468239
Ni	6.2461068	0.1239663	50.385512
Ag	5.0689042	0.1676482	30.235368
Sb	4.6249728	0.5618661	8.231450
As	3.0726933	0.8645727	3.554002
Co	1.8050047	0.6285243	2.871814
Cd	2.8622009	0.5755062	4.973362
Sn	2.0769384	0.4730687	4.390352
Te	0.6365768	0.0764396	8.327840
Bi	4.0604430	0.5006785	8.109881
Mn	3.0204249	0.3490243	8.653911
Li	0.4824261	0.3262638	1.478638
Al	1.8325815	0.5987685	3.060584
V	1.1052568	0.2861034	3.863137
Cr	0.8796267	0.2744055	3.205572
Rb	0.0861777	0.0274563	3.138717
Sr	0.3506569	0.3126728	1.121482
Ba	0.2468601	0.1080916	2.283805

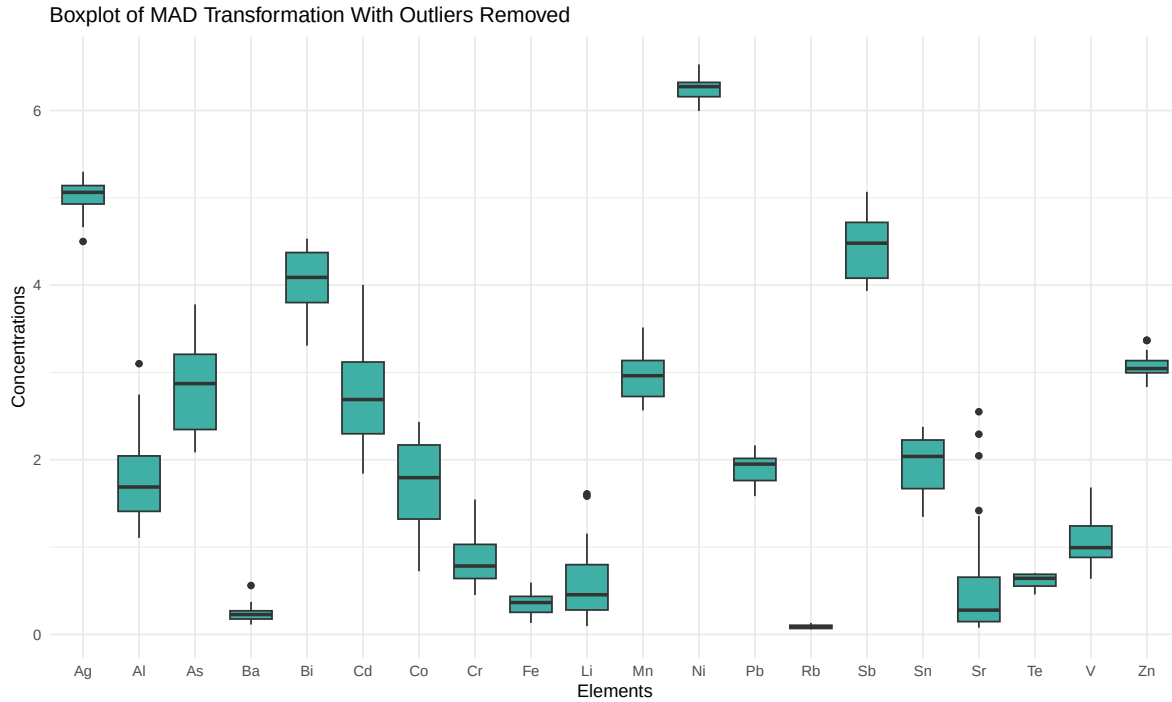
From {TABLE NUMBER} we observed some massive deviations, as expected.

Next the robust Z score was used to identify outliers that were candidates for removal. For

our cutoff, we went with Leys et al. (2013) recommendation, who argued that the MAD is the most robust scale measure under contamination and recommend a median plus or minus 2.5 time MAD rule for outlier detection.



Cu	Zn	Pb	Fe	Ni	Ag	Sb	As
4.3694479	2.9444390	0.7178398	0.5596158	6.2085900	5.0937502	3.5025499	4.0552572
Co	Cd	Sn	Te	Bi	Mn	Li	Al
0.2926696	1.7917595	1.6292405	0.6471032	3.9278964	3.4177267	0.4762342	1.8421357
V	Cr	Rb	Sr	Ba			
1.1052568	0.8197798	0.0861777	0.1570037	0.2776317			



	Zn	Pb	Fe	Ni	Ag	Sb	As	Co
3	1.214110	0.6590575	-0.5505833	1.810529	1.585719	1.539753	1.1665337	0.8100983
5	1.142787	0.4593811	-0.8249545	1.831336	1.620624	1.439718	1.0082645	0.6999087
6	1.142787	0.7012321	-1.7489684	1.850504	1.624361	1.414828	0.8029709	0.0561821
8	1.169032	0.6751417	-1.3092328	1.841834	1.625588	1.466948	0.8614114	0.3595613
9	1.079918	0.5470280	-1.0088362	1.821321	1.624361	1.547665	1.1715046	0.6083352
10	1.214110	0.6598054	-0.5703743	1.804617	1.539753	1.551491	1.1627361	0.8015255
	Cd	Sn	Te	Bi	Mn	Li	Al	
3	1.0515909	0.7135478	-0.7038114	1.224128	1.035107	0.47588500	0.8292172	
5	0.6100642	0.7711137	-0.7821427	1.408382	1.160170	0.45938112	0.9296794	
6	0.8590234	0.8133688	-0.3960925	1.419140	1.035107	0.12912558	0.5181769	
8	0.7662871	0.7308949	-0.5036581	1.407561	1.165275	0.09100400	0.9784237	
9	1.2383706	0.4658127	-0.4352494	1.509627	1.165275	-0.01051388	0.8067926	

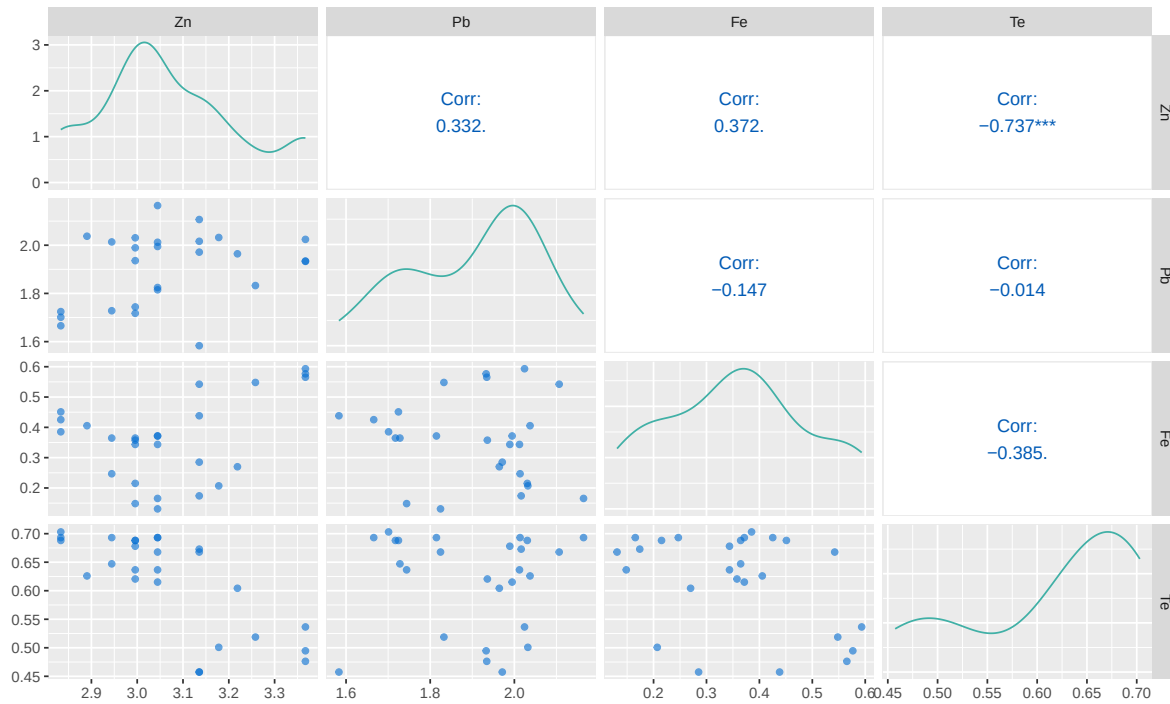
10	1.0455312	0.7103512	-0.7418456	1.209977	1.105398	-0.22064469	0.7179716
	V	Cr	Rb	Sr	Ba		
3	0.38224257	0.26875367	-2.032342	0.8296581	-1.2036242		
5	0.52039274	0.43530386	-2.177463	-0.4352494	-0.9900353		
6	0.17469888	0.06270315	-2.259909	-0.9900353	-0.5805048		
8	0.41549942	0.19222319	-2.259909	-0.1097166	-1.4648540		
9	0.26875367	0.02220262	-2.693280	-2.2599087	-1.0281321		
10	0.09404783	-0.12825761	-2.177463	-1.6574715	-1.3380214		

The QQ plot shows very skewed data which we would expected...

What is after we have “cleaned” up the outliers our $|z| > 3$ rule removes samples 13, 25, and 38 which is exactly the sample ingots that Caponetti et al. set aside as their G4 group.

{should probably explain why cutting out 11 samples will be good for the rest of our analysis, something along the line of outliers, terrible samples and low sample quality}

2.3.1 pairwise between CU, Zn, Pb, and Sn using the clean data



Also done to reflect the traditional primary components that make up orichalcum, showing the relationship between these four elements.

2.4 Multivariate normal distribution

What tests did we use? Henze Zirkler

Why did we use those tests? #ALICE STUFF

Henze_Zirkler Test Results:

	Test Statistic	p.value	Method	MVN
1 Henze-Zirkler	1.001	<0.001	asymptotic	Not normal

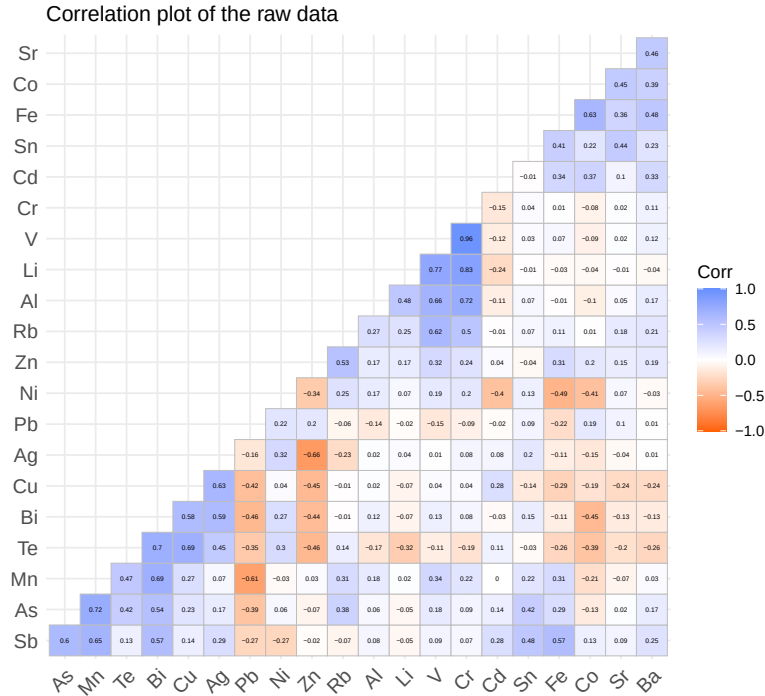
Henze_Zirkler Test Results:

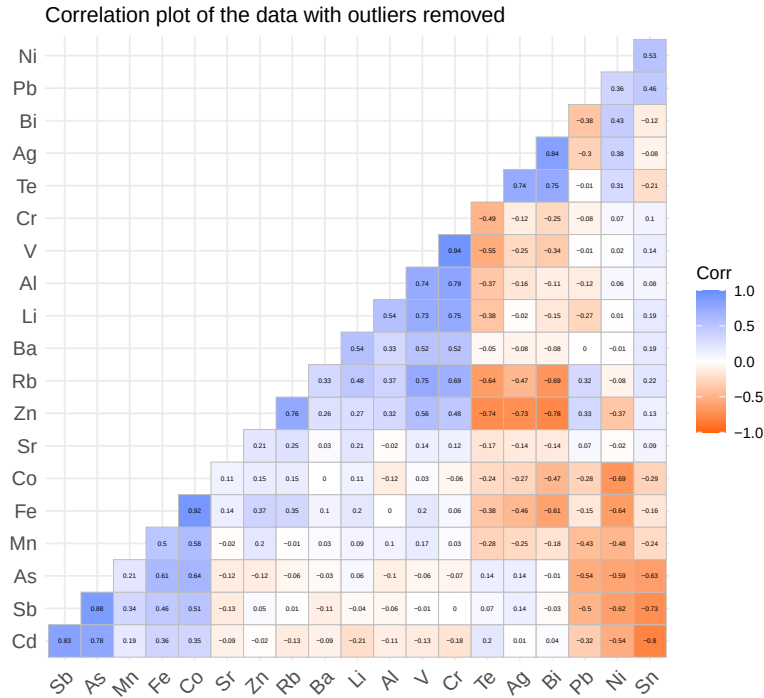
	Test Statistic	p.value	Method	MVN
1 Henze-Zirkler	1	0.055	asymptotic	Normal

Due to our Henze Zirkler test for both the un-transformed data (p-value <0.001) and the transformed data (p-value 0.046) we can safely reject the null hypotheses and conclude that the data regardless if it is transformed or not does not follow a multivariate normal distribution.

As our data isn't multivariate it isn't appropriate to perform a mahalanobis distance. {could potentially do need to research tho}

2.5 Correlation plots before and after





Seeing a lot more correlation in the data with the massive outliers removed. Correlation plot some things to highlight; is that the strong positive association between V and Cr is also strongly positively association in the data with the outlying samples removes.. The transformed data also clearly shows a strong positive association for Bi and Ag which the un-transformed data doesn't show as significantly. the negative association between Zn and Te becomes more stronger in the transformed data.

The transformation didn't just strength some of the associations it also happened to reverse the sign for several of the relationships, Bi and Mn was one the raw data had a positive association but the transformed data turned it into a negative association similar with the relation ships between Sb and Sn as well as Rb and Te. It went the other direction (negative to positive) for As and Co. This is a positive sign that the robustness of the transformation, i.e removing some of the outliers demonstrates their influence on the raw data.

Given the uncertainty about multivariate normality, a linear data analysis was chosen over a primary component analysis.

4. LDA

4.1 PAM

We started by performing a PAM (Partitioning Around Medoids) analysis

```
'data.frame':  26 obs. of  20 variables:
 $ Zn: num  2.83 3.37 3.14 3.14 3.22 ...
 $ Pb: num  1.67 1.93 1.58 2.02 1.96 ...
 $ Fe: num  0.425 0.577 0.438 0.174 0.27 ...
 $ Ni: num  6.28 6.11 6.24 6.36 6.31 ...
 $ Ag: num  5.3 4.88 5.06 5.08 5.08 ...
 $ Sb: num  4.89 4.66 4.22 4.12 4.34 ...
 $ As: num  3.62 3.21 2.74 2.23 2.37 ...
 $ Co: num  2.24 2.25 2.01 1.06 1.43 ...
 $ Cd: num  3.04 2.86 1.84 2.36 2.15 ...
 $ Sn: num  1.76 2.04 2.16 2.26 2.08 ...
 $ Te: num  0.693 0.495 0.457 0.673 0.604 ...
 $ Bi: num  4.3 3.4 4.09 4.13 4.09 ...
 $ Mn: num  3.08 2.82 3.19 2.82 3.21 ...
 $ Li: num  1.15 1.61 1.58 1.14 1.1 ...
 $ Al: num  1.7 2.29 2.53 1.68 2.66 ...
 $ V : num  0.936 1.466 1.683 1.191 1.515 ...
 $ Cr: num  0.837 1.308 1.545 1.065 1.212 ...
```

```
$ Rb: num 0.0677 0.131 0.1133 0.1044 0.1044 ...
$ Sr: num 0.14 2.293 0.647 0.372 0.896 ...
$ Ba: num 0.166 0.3 0.372 0.56 0.231 ...
```

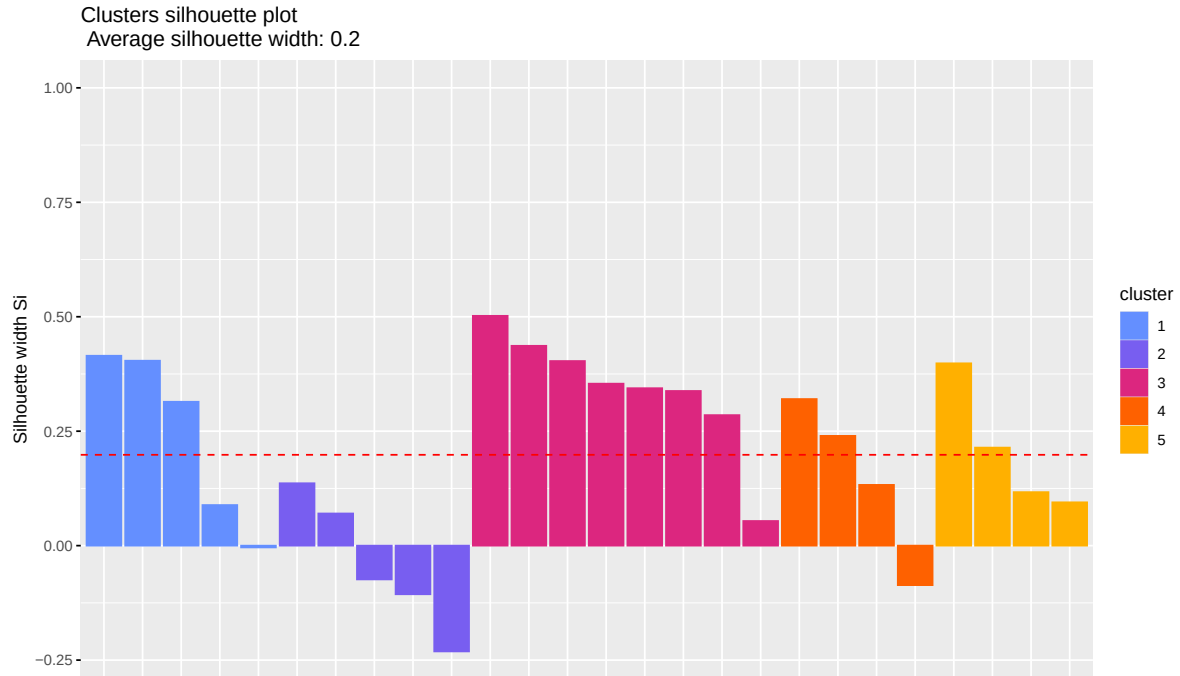
```
[1] 26 20
```

Table 8: Figure 8: Table contain the k scaled values

k	avg_silhouette
2	0.1819397
3	0.2456829
4	0.1617763
5	0.1794853
6	0.1764320
7	0.1644250
8	0.1695008
9	0.1681690
10	0.1517058
11	0.1473579
12	0.1278878
13	0.1226697
14	0.1126834
15	0.1163527
16	0.1256641
17	0.1002570
18	0.0947269
19	0.1025459

k	avg_silhouette
20	0.1096880
21	0.1068241
22	0.0987708
23	0.0931162
24	0.0656316
25	0.0348698
26	0.0000000

	cluster	size	ave.sil.width
1	1	5	0.24
2	2	5	-0.04
3	3	8	0.34
4	4	4	0.15
5	5	4	0.21



The silhouette value/metric is a measurement of the quality of a clustering. between -1 and 1. The data reports the optimal silhouette width as 5, with 0.6496032, indicating some. There are several other clustering values with a reported similar quality.

Due to the ambiguous clustering, we opted to perform more cluster analyses to determine if this was a good fit.

MANHATTAN NOT EUCLEDIAN.

3.3 Cluster Analysis

Hierarchical cluster using Manhattan Distance as a measure. Because more robust to outliers than Euclidean etc.

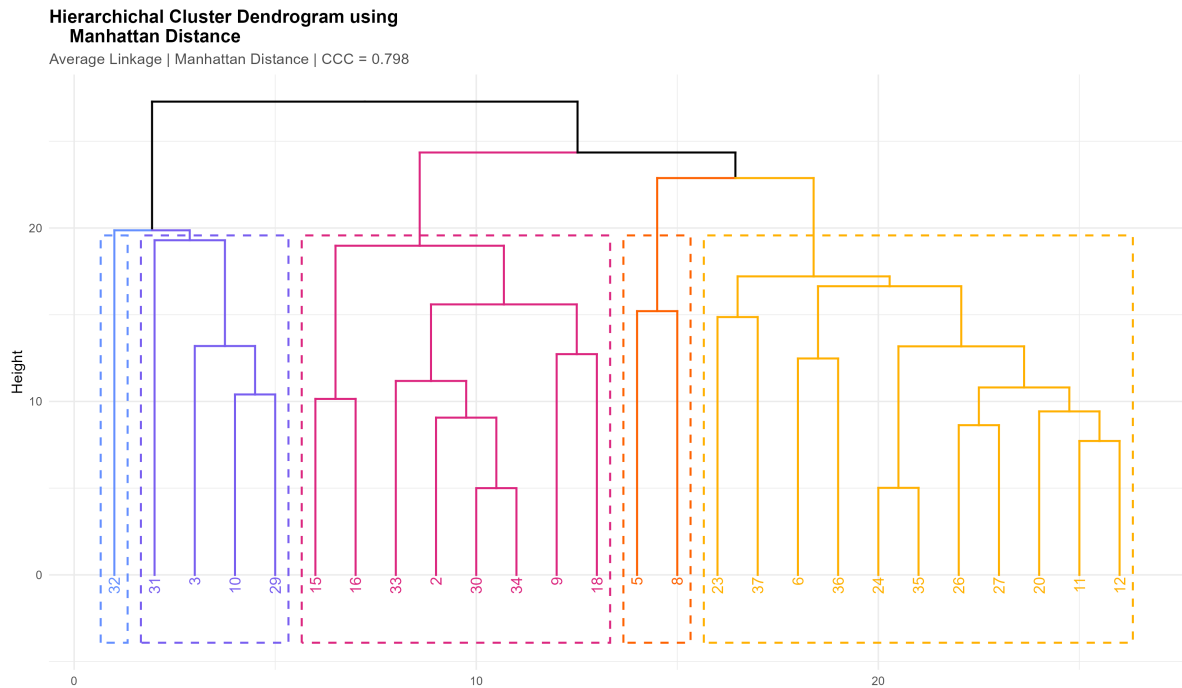


Figure 2: Dendrogram

```
d <- dist(scaled, method = "manhattan")

HiMan <- hclust(d, method = "average")

rownames(scaled) <- rownames(ICN)

p <- fviz_dend(
  HiMan,
  k = 5,
  rect = TRUE,
```

```

k_colors = my_col,
rect_border = my_col
)

head(ggplot_build(p)$data[[2]]$label)

```

```

[1] 32 31 3 10 29 15
26 Levels: 32 31 3 10 29 15 16 33 2 30 34 9 18 5 8 23 37 6 36 24 35 26 ... 12

```

```

length(unique(ggplot_build(p)$data[[2]]$label))

```

```

[1] 26

```

Something interesting top note is that for both cluster options they both cluster sample's 15, 10, and 22 this could be because of ...

Cophenetic correlation coefficient 0.825 is pretty dang good. I'm going to remake this using ggplot, should be nicer.

3.4 Fuzzy Analysis

As the third and final method of clustering, we performed a Fuzzy analysis.

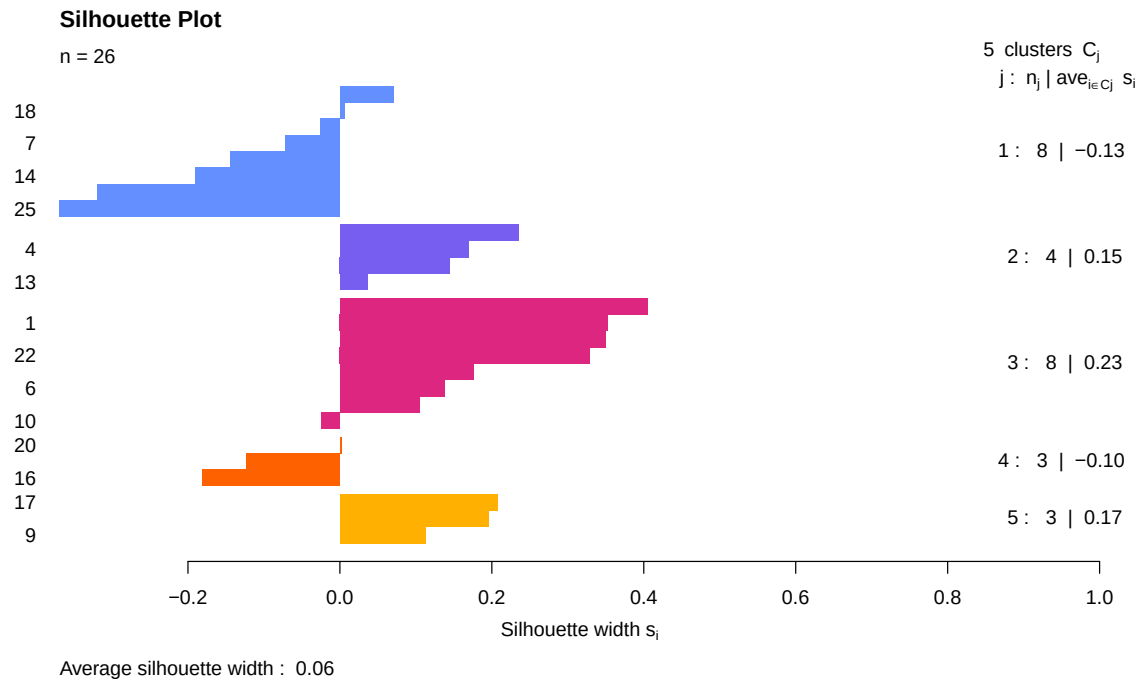
Table 9: Fuzzy K Silhouettes Clusterings

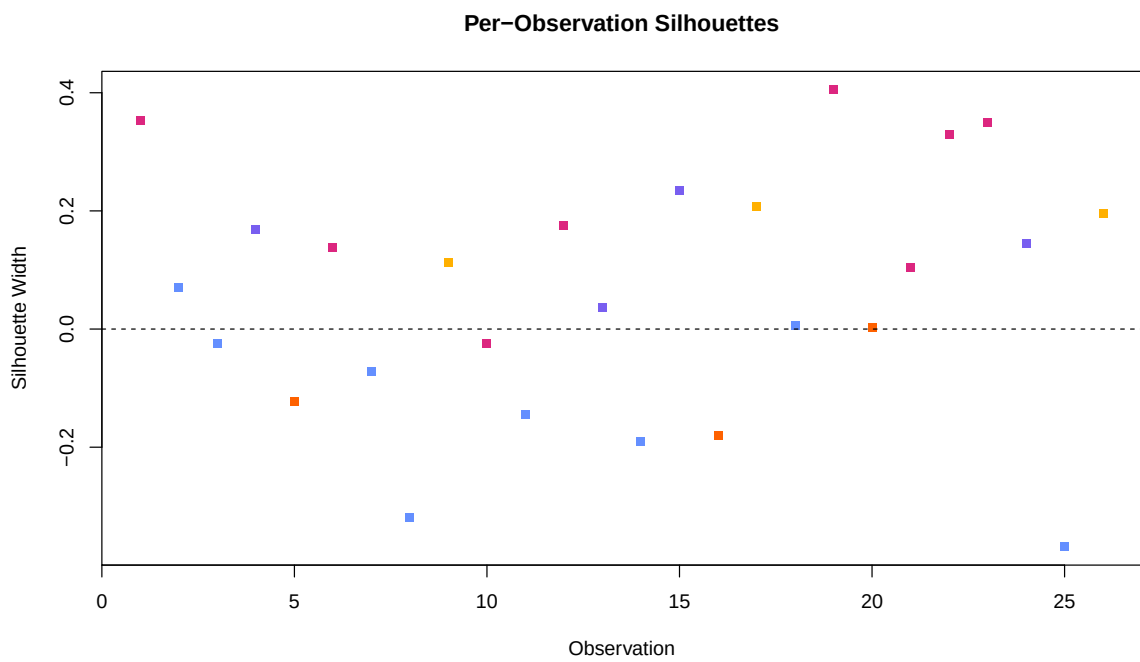
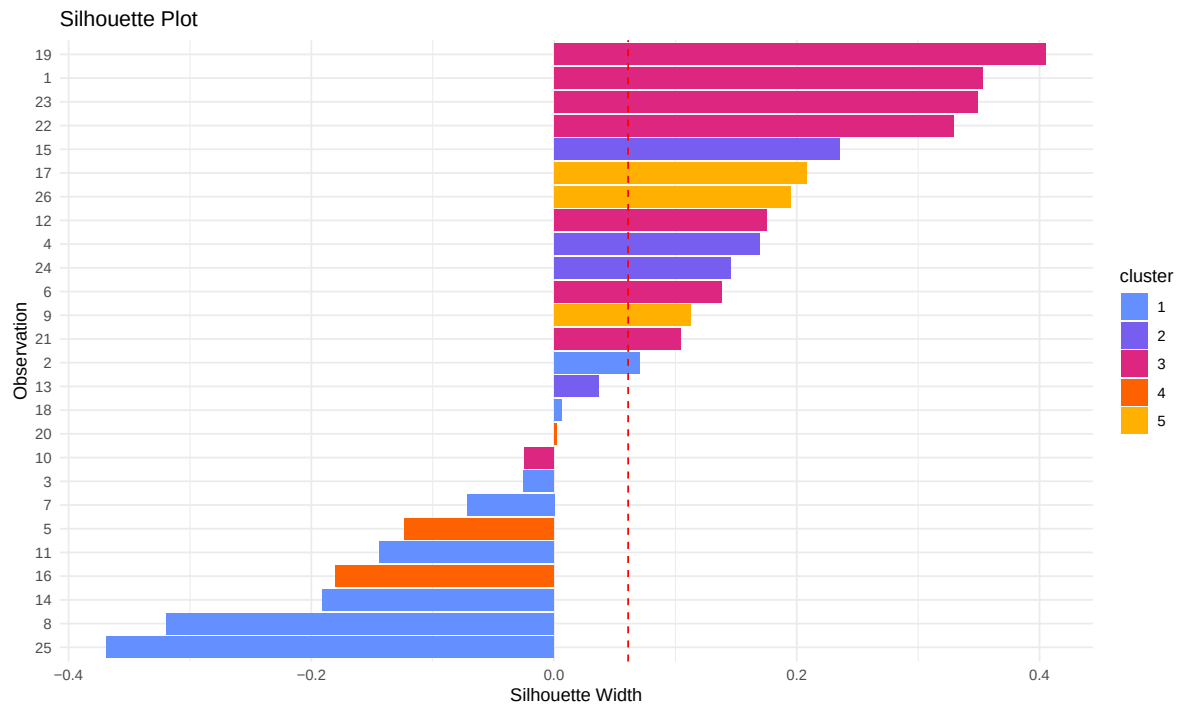
	Clus 1	Clus 2	Clus 3	Clus 4	Clus 5
2	0.1837223	0.1617960	0.3647139	0.1604535	0.1293143
3	0.2743970	0.1759881	0.1341465	0.2058822	0.2095862

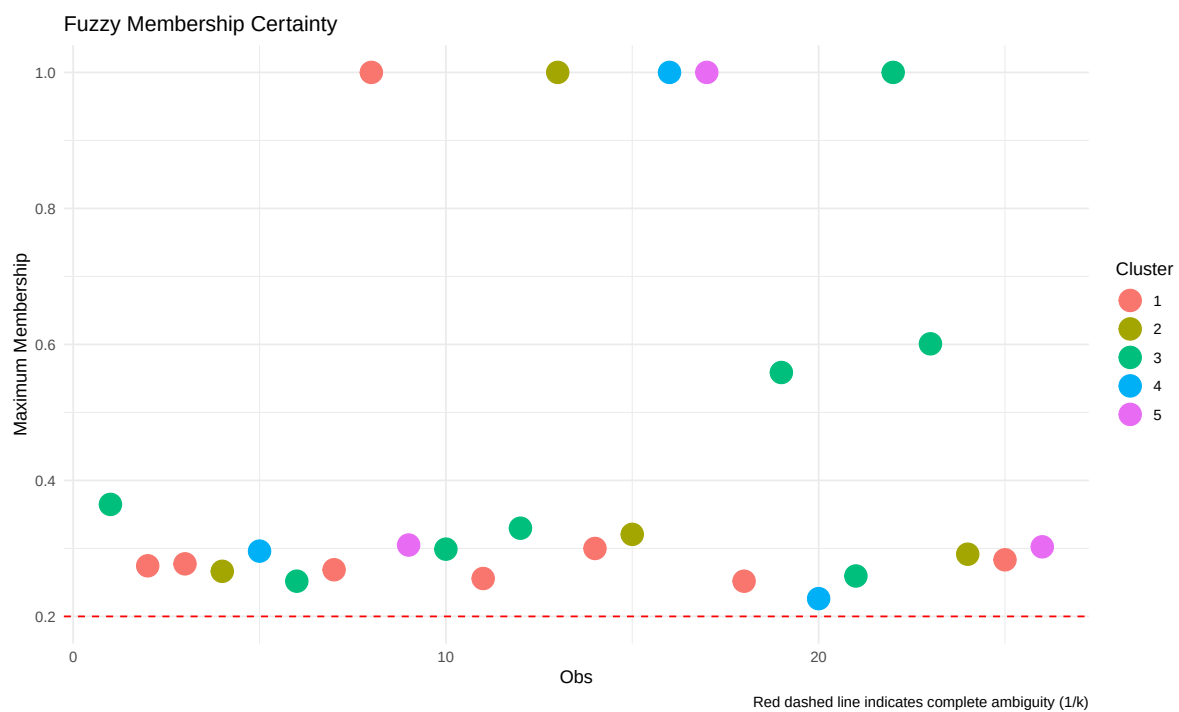
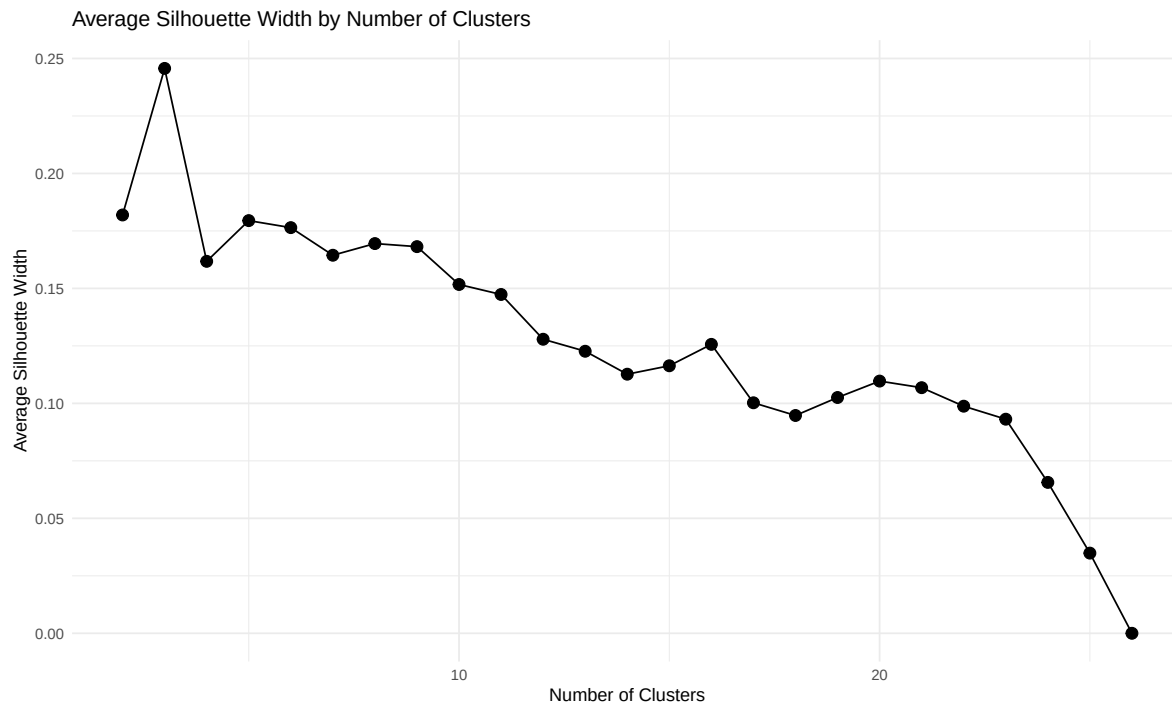
	Clus 1	Clus 2	Clus 3	Clus 4	Clus 5
5	0.2772959	0.1777119	0.1252760	0.2436443	0.1760719
6	0.2547068	0.2663341	0.1032215	0.1923250	0.1834126
8	0.2712500	0.1532005	0.1050444	0.2960191	0.1744860
9	0.2007902	0.1763241	0.2518103	0.2264072	0.1446682
10	0.2686159	0.1780810	0.1409975	0.2338997	0.1784058
11	1.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.2997819	0.2026460	0.0612202	0.1313858	0.3049661
15	0.2088894	0.1965969	0.2989064	0.1483544	0.1472529
16	0.2557350	0.1850173	0.1764975	0.2293714	0.1533788
18	0.1651782	0.1500243	0.3297245	0.1480776	0.2069954
20	0.0000000	1.0000000	0.0000000	0.0000000	0.0000000
23	0.2999084	0.1586979	0.0856721	0.2610732	0.1946484
24	0.2277548	0.3206540	0.0850383	0.1705338	0.1960191
26	0.0000000	0.0000000	0.0000000	1.0000000	0.0000000
27	0.0000000	0.0000000	0.0000000	0.0000000	1.0000000
29	0.2517924	0.1783140	0.1344882	0.2317768	0.2036286
30	0.1139200	0.1210501	0.5586861	0.1056298	0.1007140
31	0.1995135	0.1707574	0.2229763	0.2261349	0.1806178
32	0.2252052	0.1807440	0.2594404	0.1776700	0.1569404
33	0.0000000	0.0000000	1.0000000	0.0000000	0.0000000
34	0.1009872	0.1185795	0.6008686	0.0892477	0.0903170
35	0.2264733	0.2914818	0.1144725	0.1588407	0.2087317
36	0.2832469	0.2096686	0.0988054	0.2064448	0.2018344
37	0.2323875	0.1835538	0.1138111	0.1679180	0.3023296

The default value $\alpha=1$ has been set

	cluster	neighbor	sil_width
[1,]	3	2	0.35296335
[2,]	1	4	0.07047399
[3,]	1	4	-0.02521016
[4,]	2	5	0.16927964
[5,]	4	1	-0.12326565
[6,]	3	4	0.13765782







The graph of Fuzzy Membership Certainty shows that the majority of our observations are

near ambiguous in there assigned groupings.

Picks by median centrality, determines closest group and percentage it belongs in said group.

5. Results

We are unable to make direct

Caponetti, Armetta, Brusca, et al. (2017) determined that the quantity of sample was attributed to error generation. Both samples S38 and S08 had respective quantities 12.3mg and 16.0mg. The average amount of material sampled was reported at ~20mg.

The original study only had access to 38/40 ingots as they were on loan to another institute. This is an extension of that analysis, increasing the sample size at this point of the data analysis would inflate the false positive rate.

6. Discussion

Caponetti, Armetta, Martino, et al. (2017) determined 3 distinct groupings of ingots, which provides evidence of a scarce and highly prized alloy being manufactured in one place at one time. This was determined with the added variable of the physical form of the ingots.

Our results

The initial analysis was performed by non-invasive XRF, a method of getting accurate readings within a certain range of elements (find that bit again). The ICP-OE and ICP-MS chemometric techniques require a small amount of material to be sacrificed in order to obtain a full range of the elemental composition of each ingot.

The find was dated to ~600BCE, the approximate age of the ship, and by the pottery also found in the wreck.

A study on a sample group this large has never been done before. Also, this dataset is not very large. - love this sentence 10/10.

XRF is a portable, non-invasive method of getting accurate readings within a certain range of elements (find that bit again). The ICP-OE and ICP-MS chemometric techniques require a small amount of material to be sacrificed in order to obtain a full range of the elemental composition of each ingot.

Caponetti, Armetta, Brusca, et al. (2017) performed chemoanalysis to determine the trace elements of these orichalcum ingots. Using this data, they determined correlation between the observed different features and the composition to group the ingots into their appropriate cluster. This is an extension from a fledgling study by the same author, where they performed a less invasive XRF (X-Ray Fluorescence) spectroscopy of the ingots, a non-destructive test that can be performed in the field.

Their motivation was to create a starting point for giving insight into archaeometric study of the orichalcum ingots regarding their origin and manufacture technologies.

Our study also has the same motivations. We aim to provide a high level data analysis to provide a detailed, and logical, path to a linkage analysis.

Motivation, problem statement, approach, results, conclusions.

What were the limitations of this study?

What further research could be done? Different statistical analysis?

When working with museum quality pieces, that need destructive analyses performed to

Data obtained for the (NAME TABLE) containing the main component, trace element concentrations and Cu/Zn weight ratio was obtained by ICP-OES and ICP-MS. Collecting samples from 38 ancient ingots to analysis the association between the ingots elemental composition.

We also took a preliminary look at the Copper (Cu), Zinc (Zn), Lead (Pb), and Tin (Sn) as they are the four alloying elements that are known to define orichalcum and historically have been the primary elemental components used to date and pinpoint origin of other orichalcum artifacts.

7. References

- Angelucci, O. (2026). *The Sunken Ship of Gela: The Ancient Secret of the Sicilian Sea* — *medium.com*. <https://medium.com/ash-and-devotion/the-sunken-ship-of-gela-the-ancient-secret-of-the-sicilian-sea-796f56d2dcc6>.
- Caley, E. R. (1967). Investigations on the origin and manufacture of orichalcum. In *Archaeological chemistry: A symposium* (pp. 59–74). University of Pennsylvania Press. <http://www.jstor.org/stable/j.ctv512vfn.7>
- Caponetti, E., Armetta, F., Brusca, L., Chillura Martino, D., Saladino, M. L., Ridolfi, S., Chirco, G., Berrettoni, M., Conti, P., Bruno, N., & Tusa, S. (2017). A multivariate approach to the study of orichalcum ingots from the underwater gela’s archaeological site. *Microchem. J.*, 135, 163–170.
- Caponetti, E., Armetta, F., Martino, D. C., Saladino, M. L., Ridolfi, S., Chirco, G., Berrettoni, M., Conti, P., Bruno, N., & Tusa, S. (2017). *First discovery of orichalcum ingots from the remains of a 6Th century BC shipwreck near gela (sicily) seabed*.
- Frery, A. C., Gomez, L., & Medeiros, A. C. (2020). A badging system for reproducibility and replicability in remote sensing research. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, 13, 4988–4995.

- Leys, C., Ley, C., Klein, O., Bernard, P., & Licata, L. (2013). Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. *J. Exp. Soc. Psychol.*, 49(4), 764–766.
- Nichols, D. *Coloring for Colorblindness* — *davidmathlogic.com*. <https://davidmathlogic.com/colorblind/#%23%7BD%7D81%7BB%7D60-%231%7BE%7D88%7BE%7D5-%23%7BF%7D%7BF%7D%7BC%7D107-%23004%7BD%7D40-%2324e9fb>.
- Nutter, N. (2025). *Gela 1 Wreck: Archaic Greek Ship Discovery, Sewn-Plank Technology & Trade (500–480 BC)* — *nuttersworld.com*. <https://nuttersworld.com/iron-age-shipwrecks-mediterranean-sea/gela-1/>.
- Vungarala, S. K. (2023). *PCA vs LDA — No more confusion!* — *seshu8hachi*. <https://medium.com/@seshu8hachi/pca-vs-lda-no-more-confusion-fc21fb8d06e9>.